

The Fundamental Theorem of Calculus



FTC Part 2: evaluating definite integrals

- 1 Evaluate $\int_0^3 (x^2 - 2x) dx$ using the Fundamental Theorem of Calculus.
 - 2 Evaluate $\int_1^4 \sqrt{x} dx$.
 - 3 Evaluate $\int_0^{\pi/2} \cos x dx$.
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FTC Part 1: accumulation functions

- 4 Let $g(x) = \int_0^x t^2 dt$. Find $g'(x)$ using FTC Part 1.
 - 5 Let $g(x) = \int_1^x \sin(t^2) dt$. Find $g'(2)$.
 - 6 Let $g(x) = \int_0^{x^2} e^{-t} dt$. Find $g'(x)$ using FTC Part 1 combined with the Chain Rule.
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Interpreting accumulation functions

- 7 A particle's velocity is $v(t) = 3 - 2t$ (m/s). Let $s(t) = \int_0^t v(\tau) d\tau$ be its displacement from $t = 0$. Find $s(4)$ and interpret its meaning.
 - 8 Given the graph of f consists of straight-line segments with $f(0) = 0$, and $g(x) = \int_0^x f(t) dt$, explain what feature of g 's graph occurs where f crosses the x -axis, and where f has a local max.
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Average value of a function

- 9 Find the average value of $f(x) = x^2$ on $[0, 3]$.
 - 10 The temperature in a city over a 24-hour period is modeled by $T(t) = 70 + 10\sin\left(\frac{\pi t}{12}\right)$ for $0 \leq t \leq 24$. Find the average temperature over this period.
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Definite integrals with substitution-friendly integrands

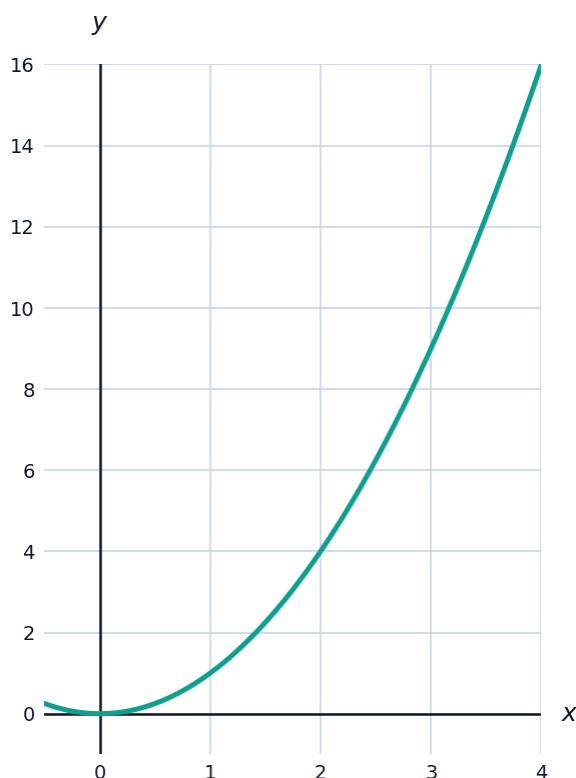
- 11 Evaluate $\int_0^1 2x(x^2 + 1)^3 dx$ using $u = x^2 + 1$.
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The Mean Value Theorem for integrals

- 12 Find the value c guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$, i.e. the point where $f(c)$ equals the average value found earlier.

Visualizing FTC Part 1 as an accumulation function

- 13 The graph shows $f(t) = t^2$ (the integrand from $g(x) = \int_0^x t^2 dt$). Explain, using the graph, why g is always increasing, and why $g'(x) = f(x) = x^2$ matches the height of the curve at each x .



A capstone FTC application

- 14 Water flows into a tank at rate $r(t) = 8 - t$ liters/minute for $0 \leq t \leq 8$ (the rate decreases as the tank fills). Using FTC, find the total volume of water added during the first 6 minutes, and explain why the rate being zero at $t = 8$ is consistent with the tank reaching capacity.

FTC Part 1 with more complex bounds

- 15 Let $g(x) = \int_3^{2x} \ln(t+1) dt$. Find $g'(x)$ using FTC Part 1 and the Chain Rule.

A second average-value application

- 16** A company's hourly production rate over an 8-hour shift is $p(t) = 50 + 10t - t^2$ units/hour, for $0 \leq t \leq 8$. Find the average production rate over the shift.